This portfolio showcases practical applications of machine learning, data visualization, and financial modelling projects developed using Python, Excel, and Google Sheets.

Detection of Fraudulent Insurance Claims Using Machine Learning

Objective

To investigate how machine learning algorithms can leverage features extracted from insurance claim datasets to detect fraudulent insurance claims and improve risk assessment accuracy.

Background

Insurance fraud detection is a classification problem characterized by class imbalance, where fraudulent claims typically represent a small proportion of total claims. This project explored multiple machine learning classifiers to evaluate their effectiveness in identifying fraudulent claims based on structured dataset features.

Approach

- Identified relevant features from the insurance claims dataset for model training.
- Conducted exploratory data analysis to understand class distribution and detect imbalance.
- Addressed the classification problem using multiple machine learning algorithms.
- Compared model performance using evaluation metrics beyond accuracy.
- Selected the best-performing model based on predictive performance.

Models evaluated included:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- XGBoost

The dataset was split into:

- 80% Training Data
- 20% Testing Data

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Model Evaluation Results

Both XGBoost and KNN achieved an overall accuracy of 94%. However, performance was further evaluated using precision, recall, F1-score, and confusion matrix analysis to assess model robustness.

Model	Precision	Recall	F1-Score	Accuracy
XGBoost	0.94	0.94	0.91	0.94
KNN	0.88	0.94	0.91	0.94

XGBoost demonstrated stronger performance in precision and better identification of true positives while maintaining high recall and overall accuracy.

Key Findings

- The dataset exhibited significant class imbalance, with fraudulent claims representing a small proportion of total observations.
- Accuracy alone was insufficient to evaluate model performance due to imbalance.
- XGBoost performed slightly better in identifying true positive fraud cases.
- Model comparison demonstrated the importance of evaluating multiple algorithms before final selection.

Tools & Technologies

- Python
- Machine Learning Libraries (Scikit-learn, XGBoost)
- Statistical Analysis
- Data Exploration Techniques

Development Approach

- Used Python to implement and train machine learning classifiers.
- Leveraged AI-assisted coding tools to generate and refine model implementation while ensuring statistical validity.
- Reviewed and validated outputs to confirm alignment with dataset characteristics and evaluation metrics.

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Learning Outcomes

- Gained practical experience in implementing supervised machine learning classifiers.
- Understood the impact of class imbalance on model evaluation.
- Learned to evaluate models using precision, recall, and F1-score rather than relying solely on accuracy.
- Developed skills in interpreting model outputs for real-world insurance applications.

Bike Sales Analysis Dashboard

Project Overview:
Performed data cleaning, transformation, and exploratory analysis on a bike sales dataset to identify key purchasing patterns across demographic and behavioral factors. Built an interactive dashboard using pivot tables and visualizations in Microsoft 365 Copilot-Excel to uncover actionable insights.

Key Data Features:
Dataset included: ID, marital status, gender, income, children, education, occupation, homeowner status, commute distance, region, age, age brackets (created), and bike purchase (Yes/No).

Data Preparation & Transformation

- Cleaned and standardized raw dataset.
- Handled missing values and structured data for analysis.
- Created a derived column — Age Brackets — by grouping age into:
  Adolescent, Middle-age, Old
- Transformed raw data into structured format for pivot table analysis.

Dashboard Visualizations

Built using Pivot Tables + Pivot Charts.

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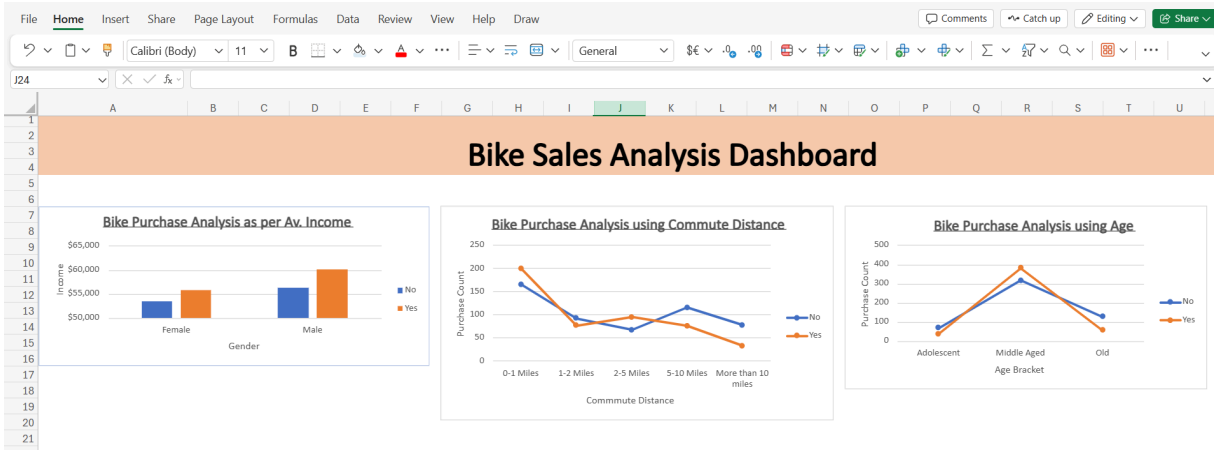

Bike Purchase vs Average Income by Gender

Insight: Males showed a higher bike purchase tendency compared to females.

Bike Purchase vs Commute Distance

Insight: Shorter commute distances correlated with higher bike purchases.

Bike Purchase vs Age Bracket

Insight: Middle-aged individuals had the highest bike purchase rate.

Tools Used
Microsoft Excel Copilot 365

Google Sheets Financial Budget Tracker

Objective

To build a structured financial tracking system for monitoring cash flow, expenses, savings, and debt using spreadsheet automation and visualization.

Key Features

- Implemented budget vs actual comparison for financial control.
- Created automated balance calculations using formulas.
- Developed interactive donut charts to visualize budget remaining actual cash balance.

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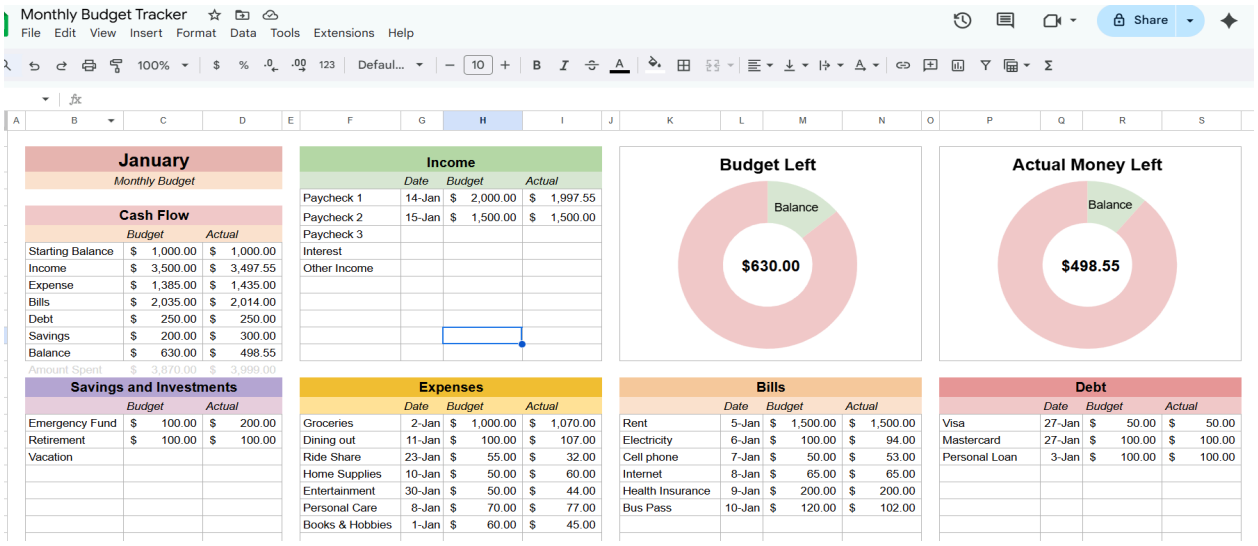

Monthly Budget Tracker

January

Monthly Budget

Cash Flow

	Budget	Actual
Starting Balance	$ 1,000.00	$ 1,000.00
Income	$ 3,500.00	$ 3,497.55
Expense	$ 1,385.00	$ 1,435.00
Bills	$ 2,035.00	$ 2,014.00
Debt	$ 250.00	$ 250.00
Savings	$ 200.00	$ 300.00
Balance	$ 630.00	$ 498.55
Amount Spent	$ 3,870.00	$ 3,999.00

Income

	Date	Budget	Actual
Paycheck 1	14-Jan	$ 2,000.00	$ 1,997.55
Paycheck 2	15-Jan	$ 1,500.00	$ 1,500.00
Paycheck 3			
Interest			
Other Income			

Budget Left

Actual Money Left

Savings and Investments

	Budget	Actual
Emergency Fund	$ 100.00	$ 200.00
Retirement	$ 100.00	$ 100.00
Vacation		

Expenses

	Date	Budget	Actual
Groceries	2-Jan	$ 1,000.00	$ 1,070.00
Dining out	11-Jan	$ 100.00	$ 107.00
Ride Share	23-Jan	$ 55.00	$ 32.00
Home Supplies	10-Jan	$ 50.00	$ 60.00
Entertainment	30-Jan	$ 50.00	$ 44.00
Personal Care	8-Jan	$ 70.00	$ 77.00
Books & Hobbies	1-Jan	$ 60.00	$ 45.00

Bills

	Date	Budget	Actual
Rent	5-Jan	$ 1,500.00	$ 1,500.00
Electricity	6-Jan	$ 100.00	$ 94.00
Cell phone	7-Jan	$ 50.00	$ 53.00
Internet	8-Jan	$ 65.00	$ 65.00
Health Insurance	9-Jan	$ 200.00	$ 200.00
Bus Pass	10-Jan	$ 120.00	$ 102.00

Debt

	Date	Budget	Actual
Visa	27-Jan	$ 50.00	$ 50.00
Mastercard	27-Jan	$ 100.00	$ 100.00
Personal Loan	3-Jan	$ 100.00	$ 100.00

Technical Skills

- Data structuring & financial modelling
- Spreadsheet formula automation
- Budget variance analysis
- Data visualization through dynamic charts
- Financial performance monitoring

Impact

- Enabled real-time tracking of income and expenditures.
- Improved visibility into cash flow management.
- Provided a structured framework for personal financial decision-making

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